

Predictors as Markets

One maker, makers in parallel, markets in series

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Abstract

There are two algebraic reasons markets appear inside learning: convex duality turns optimization into trading, and graphical factorization turns inference into networks of local markets. We develop both. For one maker, non-convexity does not obstruct coherence: no-arbitrage is a chord condition, rational flow trades the biconjugate, and a proportional fee of at least the arbitrage depth (the worst per-unit excursion of chord slopes beyond the payoff hull) restores no-arbitrage, the fee being exactly a bid-ask spread by conjugation. For makers in parallel, combining fee-bearing makers is an infimal convolution solved by one monotone clearing-price root-find with sparse fills, and the aggregate supply curve is a consolidated limit order book. For markets in series, one market per chain-rule factor makes the market an exact min-plus inference engine: parallel merge implements factor multiplication and cheapest routing implements variable elimination, both for arbitrary convex potentials. A finite-state chain of markets runs Viterbi exactly; sum-product semantics are recovered exactly on log-quadratic (Gaussian) families, where partial minimization equals marginalization up to a constant, and the market-implemented Kalman filter and exact tree marginals live on this intersection. A cost-based predictor admits a coherent market implementation if and only if its arbitrage depth is dominated by the permitted friction; friction extends the construction beyond convexity, and arbitrage enforces coherence across markets, every convex feasibility violation being a separating portfolio. The resulting object is the incentive closure of the predictor: the same operator, with every information source paying to move the state.

1 Many predictors are already markets

A cost-function market maker posts a potential C over its inventory q of outstanding shares and charges $C(q + s) - C(q)$ for a fill of s shares; prices are gradients of C , and when C is convex the standard theory applies ([Hanson 2003](#); [Abernethy et al. 2013](#)). A run of published identities makes many learning procedures markets in this literal sense. Trading against such a maker is follow-the-regularized-leader with the cost as conjugate regularizer ([Chen and Vaughan 2010](#); [Abernethy et al. 2013](#)), with trader wealth as the learning rate ([Frongillo et al. 2012](#)). A market of Kelly bettors performs Bayesian model averaging with wealths as posterior weights ([Beygelzimer et al. 2012](#)), and equilibrium prices of utility-maximizing agents implement mixtures and products of experts ([Pennock and Wellman 1997](#); [Storkey 2011](#); [Storkey et al. 2012](#); [Barbu and Lay 2012](#)).

Two further identities are used throughout. Ordinary least squares is a market of data points: each observation a quadratic maker quoting its value with capital equal to its precision (inverse variance), the estimate their merge, since merging makers is the infimal convolution of their

costs and liquidity, the inverse of price impact, adds (Bhaskara et al. 2023; Barrieu and El Karoui 2005). And a proximal step is a trade against a fee-bearing maker: the proximal operator of $f|\cdot|$ is the soft-threshold, which is the optimal response to a proportional fee (Lemma 4 below), and the prox of any convex g is the response to a maker charging g . The two canonical penalties are then the two market primitives: ridge is a zero-quoting participant with capital λ , lasso is a fee of λ , and the theorem that regularization is robustness to data perturbation (El Ghaoui and Lebrete 1997; Xu et al. 2009) acquires a market reading in which the adversary’s budget is priced rather than assumed.

We ask the converse: which predictors are markets, and what does the market form demand? The answer has two independent parts, and the paper is organized around them. Convex duality turns optimization into trading: sections 2–4 treat one maker and sections 5–7 makers in parallel. Graphical factorization turns inference into networks of local markets: sections 8–11 treat markets in series, with the precise semiring statement in section 8. Section 12 states the characterization. Throughout, a scalar security settles at $\varphi(\omega) = \omega \in [-1, 1]$; vector statements substitute the convex hull of payoff vectors. Reference implementations and numerical theorem tests accompany the paper in the mechanisms repository.

2 Coherence is a chord condition

Proposition 1 (no-arbitrage without convexity). *For a path-independent maker (one whose charge depends only on the inventory endpoints) with cost C , no convexity assumed, there is no outcome-independent profitable fill from any state if and only if every chord slope $[C(q + s) - C(q)]/s$ lies in the convex hull of payoffs: for all q, s ,*

$$C(q + s) - C(q) \geq \min_{\omega} \varphi(\omega) s \quad (\text{scalar: } C \text{ is 1-Lipschitz}).$$

Proof. A fill s from state q has sure profit $\min_{\omega} \varphi(\omega) s - [C(q + s) - C(q)]$; positivity for some pair (q, s) is exactly the failure of the displayed inequality, and the inequality for all pairs says every chord slope is supported by the hull. ■

Round trips are refunds for any C , convex or not, because the charge telescopes over closed paths. When the chord condition fails the exploit is the accumulation of sure-profit net positions at states where quotes exit the hull.

Convexity enters the standard axiomatics through *information incorporation*, the marginal-cost monotonicity condition of Abernethy et al. (2013), not through no-arbitrage. The habit of assuming convexity has bundled two different guarantees: monotone price response to flow, and absence of sure-loss opportunities. The axiom sets impose the first, convexity follows, and no-arbitrage is then derived with convexity in hand, so the possibility of a coherent maker with non-monotone quotes never arises. Proposition 1 separates the guarantees: dropping information incorporation while keeping the chord condition leaves a coherent non-convex maker, exhibited numerically in the companion repository. Arbitrage theory without convexity is developed in a different formalism by Lépinette and Tran (2017).

3 The market trades the biconjugate

Let $\hat{C}$ denote the lower convex envelope of a chord-coherent cost C (its biconjugate on the line (Rockafellar 1970)) and $g = C - \hat{C} \geq 0$ the gap.

Proposition 2 (contact and pass-through). *A myopic risk-neutral trader with believed mean μ in the hull, facing the maker at state q : (i) has optimal fills landing on the contact set $\{C = \hat{C}\}$, and (ii) earns maximal expected profit*

$$\Pi_C(q, \mu) = \Pi_{\hat{C}}(q, \mu) + g(q),$$

the envelope profit plus the gap at the starting state.

Proof. $\sup_s \mu s - [C(q + s) - C(q)] = \sup_x [\mu x - C(x)] - \mu q + C(q)$. Since $C \geq \hat{C}$ with equality on the contact set, and a maximizer of an affine function minus C is a point where an affine minorant touches C , hence touches $\hat{C}$, the supremum equals $\sup_x [\mu x - \hat{C}(x)]$ and is attained on the contact set. Adding and subtracting $\hat{C}(q)$ gives (ii). ■

The maker behaves observationally like its convex envelope. Concave stretches are unquotable intermediate states, holes in the maker’s book of standing quotes that rational flow jumps across. Whoever lands inside one (noise) overpays the gap at the landing state, and by (ii) the next rational trader recoups it, so the maker is a conduit keeping envelope differences over any rational-to-rational span, and off-contact states are transient. The trade-set analogue for constant-function market makers (CFMMs, the decentralized-exchange design) is the canonical concave trading function: an arbitrary invariant is behaviorally equivalent to a concave one (Angeris et al. 2024), with the limits of concavification mapped by Frongillo et al. (2024).

Read as beliefs, a hole is a multimodal quote: the maker takes prices on either side of the gap and refuses every price inside, jumping discontinuously as flow accumulates. This is the book-side shape of the bimodal implied densities documented around binary events (Melick and Thomas 1997; Clark and Amen 2017).

4 Frictions price the remaining failure

The failure mode surviving §2 is chord slopes exiting the hull.

Proposition 3 (fees buy bounded incoherence). *If chord slopes exit the hull by at most ε per unit, i.e. $C(q + s) - C(q) \geq \min_{\omega} \varphi(\omega)s - \varepsilon|s|$, then a proportional fee $f \geq \varepsilon$ restores no-arbitrage.*

Proof. Sure profit with the fee is $\min_{\omega} \varphi(\omega)s - \Delta C - f|s| \leq (\varepsilon - f)|s| \leq 0$. ■

This is the mechanism-level counterpart of a theorem of mathematical finance: arbitrarily small proportional transaction costs restore no-arbitrage for price processes that are arbitrageable frictionlessly, with a consistent price system inside the spread as certificate (Guasoni 2006; Guasoni et al. 2010). In prediction markets the precedent is fees sized to expected arbitrage profit restoring bounded loss under privacy noise (Cummings et al. 2016; Frongillo and Waggoner 2018). Call the smallest ε for which the hypothesis of Proposition 3 holds the cost’s *arbitrage depth*: the worst per-unit excursion of its chord slopes beyond the payoff hull, and so the minimum viable spread. A venue quoting a wide spread to cover a deeply incoherent cost is arbitrage-free but uninformative in proportion: the market prices the model’s incoherence as uncertainty.

5 A linear fee is a bid-ask spread

Why a fee at all? A cost-function maker’s charge telescopes, so round trips are free: the maker earns nothing on uninformed flow and cannot recover from volume its adverse-selection losses,

the losses to better-informed traders. That no state-dependent cost can charge a round trip, and that a path-dependent volume charge repairs it, is due to Othman and Sandholm (2012); the linear charge $f|s|$ is their volume levy in its simplest convex form, chosen because it conjugates in closed form. Write $g_q(s) = C(q+s) - C(q)$ for a convex C and $m = C'(q)$ for the marginal price.

Lemma 4 (fee–spread duality). *Let $T_f(x) = \text{sign}(x) \max(|x| - f, 0)$ denote the soft-threshold. Then the fee-bearing cost $\tilde{C}_q = g_q + f|\cdot|$ has conjugate*

$$\tilde{C}_q^*(p) = g_q^*(m + T_f(p - m)),$$

zero if and only if $|p - m| \leq f$.

Proof. The conjugate of a sum of closed proper convex functions with overlapping relative interiors of domains is the infimal convolution of the conjugates (Rockafellar 1970, Thm. 16.4), and the conjugate of $f|\cdot|$ is the indicator of $[-f, f]$. Hence $\tilde{C}_q^*(p) = \min_{|u| \leq f} g_q^*(p - u)$, the minimum of the convex function g_q^* over $[p - f, p + f]$. Since $g_q^* \geq 0$ with equality exactly at m , the minimum is attained at the projection of m onto the interval, which is $m + T_f(p - m)$. ■

The maker quotes $\text{ask} = m + f$ and $\text{bid} = m - f$ and trades nothing in between: a proportional fee is a bid-ask spread, the same duality by which a proportional transaction cost confines the pricing functional to the bid-ask band (Jouini and Kallal 1995). Because conjugation reads only biconjugates, the lemma survives non-convexity with the envelope in place of the cost:

Lemma 5 (the envelope form). *For any chord-coherent C with envelope $\hat{C}$, the fee-bearing maker's no-trade band at state q is the interval of beliefs within f of the envelope's marginal price, non-empty exactly on the contact set; at off-contact states every belief in the hull yields profit at least $g(q)$.*

Proof. By Proposition 2, profit at belief μ is the envelope profit plus $g(q)$; with the fee the envelope profit is the soft-thresholded conjugate of Lemma 4, zero on the band around $\hat{C}'(q)$. On the contact set $g(q) = 0$ and the band survives; off it the additive $g(q) > 0$ leaves no zero. ■

6 Makers in parallel

Let makers $i = 1..n$ hold inventories q_i with convex costs C_i , liquidities of their choosing, and fees f_i of their choosing; write $m_i = C'_i(q_i)$ for maker i 's marginal price and $\text{ask}_i = m_i + f_i$, $\text{bid}_i = m_i - f_i$ for its quotes, as in Lemma 4.

Lemma 6 (combination). *Let $\tilde{C} = \tilde{C}_1 \square \dots \square \tilde{C}_n$ and define each maker's supply*

$$s_i(p) = \begin{cases} (C'_i)^{-1}(p - f_i) - q_i, & p \geq \text{ask}_i, \\ 0, & \text{bid}_i < p < \text{ask}_i, \\ (C'_i)^{-1}(p + f_i) - q_i, & p \leq \text{bid}_i, \end{cases}$$

each non-decreasing in p . Fix a demand Δ for which a clearing price p^ with $\sum_i s_i(p^*) = \Delta$ exists. Then: (i) $\tilde{C}^* = \sum_i \tilde{C}_i^*$, a sum of soft-thresholded profit functions; (ii) the split $s_i = s_i(p^*)$ attains $\tilde{C}(\Delta)$, and any optimal split satisfies $C'_i(q_i + s_i) + f_i \text{sign}(s_i) = p^*$ for $s_i \neq 0$ and $|p^* - m_i| \leq f_i$ for $s_i = 0$; (iii) the split is sparse: every maker whose quote band strictly contains p^* trades exactly zero.*

Proof. (i) is the conjugate-sum identity applied to the convolution (Rockafellar 1970). For (ii), the split is feasible by choice of p^* , and $p^* \in \partial \tilde{C}_i(s_i(p^*))$ for every i : when $s_i(p^*) \neq 0$ the subgradient is $C'_i(q_i + s_i) + f_i \text{sign}(s_i) = p^*$, and when $s_i(p^*) = 0$ it is the interval $[m_i - f_i, m_i + f_i] \ni p^*$. A common multiplier certifying every coordinate is exactly the optimality condition for $\min\{\sum_i \tilde{C}_i(s_i) : \sum_i s_i = \Delta\}$. (iii) restates the zero branch. ■

The computation is a scalar monotone root-find whatever n is, and the fee costs nothing beyond a horizontal shift of each supply curve. The $|s|$ terms are an ℓ_1 penalty, so sparsity arrives for the same reason it does in the lasso, and for the same reason proportional transaction costs produce no-trade regions and sparse portfolios (Olivares-Nadal and DeMiguel 2018). A small trade routes entirely to the tightest quote; a growing trade pushes that maker's fee-adjusted marginal price through the next band and spills over, consuming makers in fee order.

Corollary 7 (zero fees). *With $f_i \equiv 0$ the convolution reduces to the fee-free merge: conjugate regularisers add, and for a perspective family $C_b(q) = b C_1(q/b)$ liquidity adds, $C_{b_1} \square C_{b_2} = C_{b_1+b_2}$ (Bhaskara et al. 2023). For quadratic makers the merge is Gaussian fusion: the merged quote is the precision-weighted mean of the makers' quotes, and the merged liquidity is the sum of their liquidities (precisions).*

Corollary 8 (the order book). *The aggregate supply $S(p) = \sum_i s_i(p)$ is non-decreasing, identically zero on $(\max_i \text{bid}_i, \min_i \text{ask}_i)$ when that interval is non-empty, flat wherever every maker's band covers p , and smooth and strictly increasing wherever some maker is active. Read as a market: best bid and ask are the tightest quotes, depth at each price is the sum of the active makers' closed-form supplies, and large orders walk the levels. The aggregate of linear-fee makers is a consolidated limit order book, and in producer-theory terms Lemma 6 is Marshall's horizontal summation of firm supply curves (Marshall 1890; Mas-Colell et al. 1995) with the reversibility of share production patched by the fee. The economics of the book assembled from competing liquidity suppliers is Glosten (1994), with convergence of strategic schedules in Biais et al. (2000).*

Proposition 9 (a deep co-quoter is Moreau smoothing). *Merging a maker with cost C with a quadratic co-quoter of cost $s^2/(2\lambda)$ yields the venue with cost the Moreau envelope $e_\lambda C(x) = \min_y C(y) + (x - y)^2/(2\lambda)$, a pointwise minimum of the convex branches $y \mapsto C(y) + (\cdot - y)^2/(2\lambda)$, two of which can attain the minimum at the same x (a crossing). Then: (i) coherence is preserved: if C is chord-coherent, so is $e_\lambda C$ for every λ ; (ii) the convexity defect of $e_\lambda C$, the largest slope drop at a crossing, equals the separation of the competing minimizers divided by λ , so if all competing minimizers lie in an interval of diameter D (as when the set where C exceeds its convex envelope is bounded and C is convex outside it), the defect is at most D/λ ; (iii) the symmetric double well attains the bound at every depth.*

Proof. The merge is infimal convolution, and inf-convolution with the quadratic is the Moreau envelope (Rockafellar 1970). For (i), wherever the infimum is attained at $y^*(x)$ with C differentiable there, the first-order condition gives $e_\lambda C'(x) = (x - y^*)/\lambda = C'(y^*)$: the envelope's derivative is a selection of C 's derivatives, chords average derivatives, and C 's derivatives lie in the payoff hull by hypothesis. For (ii), at a crossing x_0 with competing minimizers $y_1^* < y_2^*$ the branch slopes are $(x_0 - y_i^*)/\lambda$, so the drop is exactly $(y_2^* - y_1^*)/\lambda$. ■

Part (i) corrects a possible misreading of the construction: the merge never creates arbitrage, at any depth, so what the deep co-quoter buys is expressiveness. The holes in the book close at rate $1/\lambda$, and the residual kink of the symmetric double well is priced by a fee of the same order. Here λ is the co-quoter's liquidity: its price impact is $1/\lambda$, so large λ means a deep book, not a strong pull.

The two repairs are the two market primitives again: friction (ℓ_1 , the fee) and participant

depth (ℓ_2 , capital), lasso and ridge. The identification maps the penalty terms, not the statistical procedures; it is a dictionary of primitives, not an equivalence of estimators.

7 Self-set fees and adaptivity

Nothing requires the fees to be administered. Each maker may quote its own f_i : a quote inside the aggregate spread earns nothing, a quote too tight is picked off by informed flow, and the undercutting happens inside the same minimisation that clears the trade, since the inf in the convolution is a minimum over quotes. The surviving fee has the interpretation of the competitive adverse-selection charge of classical microstructure (Glosten and Milgrom 1985; Biais et al. 2000), reached through routing rather than a dealer game; this is an argument about discipline, and the strategic equilibrium of the quote game is not solved here.

Optimization already pays these frictions: a proximal step charges $\|\Delta\theta\|^2/(2\eta)$ per move, a trust region is an infinite fee outside a band, weight decay is a zero-quoting participant. In market language the stabilizers of non-convex training are the market's two repairs: friction prices chord excursions (Proposition 3) and depth fills holes (Proposition 9); non-convexity by itself is not exploitability (§2). Path-dependent optimizers correspond to makers whose quotes depend on flow history, the adaptive-liquidity territory where path independence is deliberately traded away: no maker combines path independence, translation invariance (complete-bundle prices summing to the sure payoff) and liquidity sensitivity (Othman et al. 2013), the adaptive class is axiomatized by Li and Wortman Vaughan (2013) with the homogeneous-risk-measure characterization in Othman and Sandholm (2011), no trade-history maker achieves every desideratum at once (Abernethy et al. 2014), and liquidity selection itself can be run as online learning (Nueve et al. 2026; Nueve and Waggoner 2025).

8 Two operators, one semiring

Sections 5–7 composed makers on one quantity. A model is a factorization,

$$P(\text{everything}) = \prod_{\text{nodes}} P(x_i \mid \text{parents}(x_i)),$$

and the remaining operator runs one market per factor, each pricing its conditional given what is upstream. The count of two is structural. Message-passing inference is an algorithm over a commutative semiring (Aji and McEliece 2000): one operation combines evidence about a variable (the product), one moves evidence between variables (the sum), and sum-product, max-product and min-sum are the one algorithm over different semirings.

The correspondence with the market algebra is exact once the objects are fixed, and the fixing matters: the two compositions live on the two sides of the Legendre transform. Assign each factor its potential, the negative log-density $\varphi_i = -\log P(x_i \mid \text{parents})$. For parallel composition, identify each maker's factor potential with its *conjugate*, $\varphi_i = C_i^*$: infimal convolution of costs in inventory space is addition of potentials in price space. For serial composition, the stage potentials are the *primal* leg costs of a route. The dictionary is

factor algebra	$\longleftrightarrow$	market algebra
density p_i	$\longleftrightarrow$	potential $\varphi_i = -\log p_i$
multiply factors	$\longleftrightarrow$	parallel merge: $\varphi_i = C_i^*$ add as costs inf-convolve
eliminate a variable	$\longleftrightarrow$	route through its market: inf over the shared leg

with the two sides related by the Legendre transform, which is the Laplace transform of the min-plus semiring (Litvinov 2005). The parallel law is a commuting square,

$$\begin{array}{ccc}
C_1, C_2 & \xrightarrow{\square} & C_1 \square C_2 \\
\downarrow * & & \downarrow * \\
C_1^*, C_2^* & \xrightarrow{+} & C_1^* + C_2^*
\end{array}$$

while serial composition stays primal, obeying the composition law $(\varphi_2 \circ \varphi_1)(x, z) = \inf_y [\varphi_1(x, y) + \varphi_2(y, z)]$, the min-plus product of kernels (two-argument potentials). The Legendre transform is not a computational trick here but the change of representation between the two composition laws.

Proposition 10 (the market computes min-plus inference). (i) *Parallel: merging makers multiplies factors, since $(C_1 \square C_2)^* = C_1^* + C_2^*$ and potentials add exactly when densities multiply.* (ii) *Serial: the cheapest route to a terminal position through the chain of stage markets prices the min-plus elimination $\inf_{z_1, \dots, z_{n-1}} \sum_i \varphi_i(z_{i-1}, z_i)$, the min-plus product of the stage kernels.* (iii) *On log-quadratic (Gaussian) families, partial minimization of a potential equals its marginalization up to an additive constant independent of the retained variables, so the market's min-plus messages are the sum-product messages and the market computes exact Bayesian inference.*

Proof. (i) is the conjugate-sum identity together with $-\log(p_1 p_2) = \varphi_1 + \varphi_2$. (ii) is the definition of the cheapest route: the trader chooses intermediate exposures to minimize the sum of stage costs. For (iii), write a jointly quadratic potential $q(x, y)$ with positive definite y -block Q_{yy} ; completing the square, $-\log \int e^{-q(x, y)} dy = \min_y q(x, y) + \frac{1}{2} \log \det(Q_{yy}/2\pi)$, the Schur-complement identity, and the constant does not depend on x . ■

The proposition divides its labor as follows. The min-plus identities of (i) and (ii) are algebraic and hold for arbitrary potentials; the generalized distributive law needs no convexity (Aji and McEliece 2000). What the market adds is implementability, and the condition is the one §2 identified, stated dually: a kernel is tradeable as a path-independent maker when its cost's chord slopes respect the payoff hull, equivalently when the potential $\varphi = C^*$ has effective domain inside the hull, so that $C = \varphi^*$ has subgradients in the hull. Convexity is the separate, further property of monotone information incorporation, and it is what gives the clean dual representation, not coherence. Under the implementability condition the min-plus computation is carried out by self-interested trading rather than by a solver.

Corollary (the serial algebra runs Viterbi). *Take finite state spaces and stage potentials $\varphi_t(i, j) = -\log P(X_t = j \mid X_{t-1} = i) - \log P(y_t \mid X_t = j)$. The cheapest route of Proposition 10(ii) is $\min_{x_{1:T}} \sum_t \varphi_t(x_{t-1}, x_t)$: the Viterbi decoding of the hidden Markov model, exactly, with no Gaussian structure anywhere. The statement is about the serial algebra; a coherent cost-function implementation of arbitrary finite-state kernels is not claimed. The three worked examples of this paper now form a progression: least squares is parallel composition, Viterbi is serial min-plus composition, and the Kalman filter of §10 is the Gaussian intersection where the serial computation is also Bayesian.*

Principle (the Gaussian intersection). *Log-quadratic families are exactly where probabilistic inference and market optimization coincide: a market whose traders optimize computes min-plus, an inference engine integrates, and Proposition 10(iii) says the two agree on this family up to constants that cancel in every price.* The Kalman and tree-marginal propositions below are instances of the intersection, not evidence for a universal serial thesis. Away from log-quadratics the market prices the max-product (MAP) posterior rather than the sum-product marginal, and the discrepancy is the Laplace-approximation gap of the potential. Risk aversion is an exponential tilt and plausibly interpolates between the two semirings; §12 poses this as the open question of which inference algorithm a risk-averse market runs.

9 One market per factor

The locality that makes the factorization tradable is Hanson’s modularity: in a combinatorial logarithmic market scoring rule (LMSR) a bet on $A \mid B$ moves that conditional and provably nothing else, uniquely among market scoring rules (Hanson 2007, 2003). Hanson runs one joint market over the product space; the substance of pricing it is probabilistic inference, which is why exact pricing is #P-hard (Chen, Fortnow, et al. 2008), why tournament markets price by Bayes-net inference (Chen, Goel, et al. 2008), why a deployed combinatorial market ran its price and asset updates on the junction-tree algorithm (Sun et al. 2012), and why approximate designs price over the marginal polytope (Dudík et al. 2012; Dudík et al. 2021). Securities structured by a Bayes-net factorization appear in Pennock and Wellman (2000), with the trades that preserve the structure characterized by Xia and Pennock (2011).

The serial architecture takes the factors as separate venues rather than one joint book. Market i opens on $P(x_i \mid \text{parents})$ when its conditioning information freezes, that is, when the upstream quantities it conditions on stop changing; it conditions on upstream quotes while upstream is live, and re-references when upstream settles. Settlements cascade through the directed acyclic graph like dataflow, and the schedule of markets is an unrolling of the graph. Observable nodes carry settled markets; a latent node either settles through its observable footprint, a market on a hidden state being a market on a functional of future observables scored through the model, or remains indicated but unsettled, priced by whoever warehouses the model risk.

10 The market step is the Kalman update

Take the linear-Gaussian chain $x_{t+1} = ax_t + \varepsilon_t$, $y_t = x_t + \eta_t$, and alternate two steps: a *model step*, propagating the current belief through the dynamics in mean-precision form, and a *market step*, merging the propagated belief with an observation maker quoting y_t whose capital is the observation precision.

Proposition 11 (the market step is the Kalman update). *The alternation is the Kalman filter (Kalman 1960): after each observation the market state equals the filtered mean and variance exactly.*

Proof. The model step is the standard prediction of mean and variance. Merging quadratic makers adds precisions and precision-weights means (the fusion form of Corollary 7), which is the information-form measurement update; the covariance-form update follows by the usual algebra. ■

The prediction step is computed and the correction step is traded, with capital setting the weight of the data. The nearest published object is the Bayesian market maker of Brahma et

al. (2012), whose trade update is a scalar Gaussian measurement update with covariance inflation for jumps; we find no filtering reading, and no alternation with a dynamic model, in the literature.

11 Trees and loops

On a Gaussian chain with an observation at every node, the posterior at an interior node is the merge of three sources: the forward message (the filter of §10 run up to the node), the local observation, and the backward message (downstream observations pulled back through the dynamics, each pull-back a reparametrization and each combination a precision addition).

Proposition 12 (trades as messages, exact on trees). *All three messages are market operations, and the merged posterior equals the brute-force conditioning of the joint at every node: Gaussian belief propagation (Pearl 1988) with trades as messages.*

The repository verifies the identity to ten digits. The contrast with the deployed combinatorial engines is architectural: there the junction tree is the pricing algorithm inside one joint market (Sun et al. 2012); here the messages pass between venues, and the graph’s edges are market boundaries.

On loopy graphs belief propagation double-counts and its fixed points drift. In a market the same inconsistency is self-punishing.

Proposition 13 (quote inconsistency is a sure profit). *Let x be a vector of unrestricted real random variables and let separate markets quote securities paying the products $x_i x_j$ for every pair, the diagonal quoted at one; assemble the quotes into a matrix P with unit diagonal. The quotes admit a joint distribution iff $P \succeq 0$ (a Gaussian with covariance P witnesses sufficiency). Otherwise, with w the eigenvector of a negative eigenvalue $\lambda_{\min}$, the bundle with weights $w w^\top$ has price $w^\top P w = \lambda_{\min} < 0$ and payoff $(w^\top x)^2 \geq 0$: a sure profit of at least $|\lambda_{\min}|$ per unit. On three coordinates the quoted pairs are exactly the edges of a triangle, so the loopy reading is literal.*

This is the second-moment coherence condition, prices of products must form a PSD matrix (d’Aspremont 2005), read as cycle consistency. Arbitrage-enforced coherence is an old idea. Coherence is no-arbitrage (Nau and McCardle 1991); Pennock and Wellman (1996) built arbitrageur agents enforcing the additivity identities of a Bayes-net economy, with equilibrium prices equal to the network’s probabilities and distributed bidding as distributed inference; the combinatorial literature removes the arbitrage algorithmically over the marginal polytope (Kroer et al. 2016); and Saguillo et al. (2025) measure Polymarket arbitrageurs extracting \$39.6M enforcing logical coherence across markets.

Two boundaries of the proposition mark the general picture. If only a subgraph of pairs is quoted, the realizability condition is not positive semidefiniteness of a filled-in matrix but PSD-completeness of the partial one (Grone et al. 1984), and the arbitrage bundle must be supported on quoted pairs. If the coordinates are constrained, say binary $x_i \in \{\pm 1\}$, the feasible quote set is the cut polytope, a strict subset of the PSD body (Deza and Laurent 1997), so quotes can be positive semidefinite yet jointly unrealizable. In every case the coherent set is the convex hull of attainable payoffs, and Proposition 14 below is the universal form.

The scope of Proposition 13 is in any case one class of consistency failure: locally quoted beliefs that cannot be embedded in any joint distribution, detected here at second moments. Not every loopy-propagation error takes this form, but the class it covers is, in a market, free money, and the flow that harvests it pushes the quotes back toward the cone. Whether the corrected fixed point is the true marginal, a surrogate of the kind loopy propagation converges

to, or something the liquidity profile selects is open.

The PSD cone is not special. Coherent price sets are convex (they are convex hulls of payoff vectors, or their conic images), and separation turns every exterior point into a trade:

Proposition 14 (arbitrage is separation). *Let K be the closed convex hull of the attainable payoff vectors (or its image under a linear security map), so that $\inf_{z \in K} \langle y, z \rangle \leq \langle y, \varphi(\omega) \rangle$ for every outcome ω and portfolio y . If a quoted configuration $x \notin K$, any separating functional y with $\langle y, x \rangle < \inf_{z \in K} \langle y, z \rangle$ is a portfolio whose price is strictly less than its realized payoff in every state: a sure profit. Proposition 13 is the instance $K = \{P \succeq 0\}$ with the separating functional $ww^\top$.*

The proof is the separating-hyperplane theorem read as a trade (Nau and McCardle 1991; d’Aspremont 2005); the payoff-hull hypothesis is what upgrades the separation certificate to an arbitrage, and for an abstract consistency set the construction yields the certificate only. In an ordinary numerical method an infeasibility is a residual to be driven down by the algorithm; in a market it is a payoff, and whoever finds it is paid to act as the separation oracle. Arbitrageurs are decentralized separation oracles, and the friction of §4 sets the tolerance below which infeasibility is allowed to persist.

12 Closing

The characterization is for the cost-based class. Call a predictor *cost-based* if it is specified by a path-independent potential C over a security inventory, and call friction of size f *permitted* if the mechanism may charge up to f per unit traded. A cost-based predictor admits a coherent market implementation if and only if its arbitrage depth is dominated by the permitted friction; chord coherence is the $f = 0$ special case, and convexity is the separate, further property of monotone information incorporation. Three words are kept apart throughout: *coherence* (chord slopes stay in the payoff hull), *convexity* (marginal prices respond monotonically to flow), and *expressiveness* (the quoting range, which non-convexity restricts to the contact set). Within the coherent class, non-convexity spends expressiveness and spends nothing else against rational flow.

The construction throughout is one operation: *marketization*. Given an operator T from inputs to outputs, a marketization is a mechanism whose clearing computes T ; whose local contributions compose, in parallel and in series, to compute composite operators; in which inconsistent contributions create exploitable trades (Proposition 14); in which friction bounds how much inconsistency can persist (Proposition 3); and in which every participant pays to perturb the computation. One estimator becomes one maker, combining evidence becomes parallel composition, composing conditional operators becomes serial composition. A statistical procedure specifies how information would be combined if supplied honestly; its marketization implements the same operator against self-interested sources. In this sense a market is the incentive closure of a predictor: the computation, plus the dual certificates of its constraints, plus payment to whoever holds one. Duality alone supplies the middle column of

computation	certificate	payment
primal solve	dual multiplier	none
verified computation	proof of violation	none or fixed
incentive closure	separating portfolio	the certificate’s value

and the closure adds the third: a marketized computation is self-policing in proportion to the value of its own errors, with friction the exchange rate between tolerated error and the cost

of participation.

The definition has neighbors in four literatures, and each holds one column. Markets that compute are old: a combinatorial auction cleared airport slots by integer programming (Rassenti et al. 1982), electricity markets pay the dual variables of the dispatch program as nodal prices (Schweppe et al. 1988), and market-oriented programming solved distributed allocation by computing competitive equilibria of artificial economies (Wellman 1993). Algorithmic mechanism design asks when optimization is compatible with incentives (Nisan and Ronen 2001), and the classification of what can be computed incentive-compatibly was posed as a program by Feigenbaum and Shenker (2002). The middle column carries a lower bound: any communication protocol realizing an efficient allocation must reveal supporting prices (Nisan and Segal 2006), so certificates are not optional. The third column exists in practice without the duality: fraud-proof systems pay whoever exhibits an invalid state transition (Teutsch and Reitwießner 2019; Kalodner et al. 2018), for discrete transitions rather than convex programs. And the friction bound has an incentive-free ancestor: the auction algorithm’s ε -complementary slackness permits suboptimality at most $n\varepsilon$, vanishing for $\varepsilon < 1/n$ on integer data (Bertsekas 1992); Proposition 3 is the same shape of bound with the ε charged to traders. The connective tissue, the closure operation itself, serial composition of mechanisms, and the identification of violation certificates with separating portfolios, appears unoccupied.

Nothing in the definition mentions prediction. The serial law is the min-plus kernel composition of dynamic programming, control, and shortest paths; the parallel law is the additive combination of local potentials; and the closure of the whole construction is a boundary statement:

Proposition 15 (the effective boundary maker). *Let a network of makers have boundary variables b , internal variables h , and local costs C_e , each jointly convex, and define $C_{\text{eff}}(b) = \inf_h \sum_e C_e(b, h)$. Then C_{eff} is convex; the elimination may be performed variable by variable in any order; for quadratic costs each single-variable elimination is a Schur complement; and on a chain the computation is the dynamic program of Proposition 10(ii). A network of local makers is one effective maker at its boundary.*

Proof. Partial minimization of a jointly convex function is convex, and iterated infima may be taken in any order (Rockafellar 1970); the quadratic case is the completion of the square in the proof of Proposition 10(iii). ■

This is the operation that reduces resistor networks to terminal impedances, eliminates latent Gaussian variables, and takes Schur complements: internal competition disappears into an external price law, as internal nodes disappear into a Dirichlet-to-Neumann map. One caution attaches to “any order”: the effective boundary market is elimination-order invariant, but the computational cost of producing it is not, and the blowup of intermediate scopes is the treewidth phenomenon of variable elimination. Prediction is the application in which beliefs and prices share units. We leave the general theory, including the categorical formulation in which markets are min-plus kernels composed by shared inventory, outside this paper’s scope.

One principle deserves separation from the open problems it generates, because it qualifies everything above.

Principle (accessible arbitrage). *Market coherence is relative not only to friction but to the cost of discovering violations. Let $\mathcal{A}$ be a class of admissible certificate-search procedures, and for a quoted configuration x let $V(A(x); x)$ be the value of the separating portfolio a procedure A finds and $C_A(x)$ its cost. The accessible-coherence set relative to $\mathcal{A}$ is*

$$K_{\mathcal{A}} = \left\{ x : \sup_{A \in \mathcal{A}} \mathbb{E}[V(A(x); x) - C_A(x)] \leq 0 \right\}.$$

A violation persists whenever every available separation procedure costs at least as much as the certificate it finds is worth.

The supremum ranges over procedures, not certificates: two procedures can find the same portfolio at very different costs, so coherence is relative to a class of arbitrageurs. Instantiating $\mathcal{A}$ gives polynomial-time coherence, bounded-budget coherence, and latency-constrained coherence, and a venue can be arbitrage-free to ordinary participants while arbitrageable to a better-equipped class. Three nested notions of coherence result: exact (no sure-profit chord, Proposition 1), frictional (violations below the spread survive, Proposition 3), and accessible (violations too expensive to discover survive). Proposition 14 addresses the existence of arbitrage; whether a procedure in $\mathcal{A}$ can profitably prove $x \notin K$ is its accessibility, a different property. The self-policing of the marketization table is qualified accordingly: effective self-correction is the error's value net of discovery cost and friction.

Open problems, in rough order of tractability:

Which inference algorithm does a risk-averse market run? Proposition 10 says a risk-neutral market computes min-plus exactly and sum-product only on the Gaussian intersection. Risk aversion is an exponential tilt, suggesting it interpolates between the two semirings; whether a risk-averse trading population prices the marginal, the mode, or a temperature in between would complete the correspondence.

Market representations of general prediction maps. The characterization above is confined to cost-based predictors. Define what it means for an arbitrary prediction map T , from data sets to forecasts, to possess a market representation, and give conditions on T equivalent to existence. A first conjecture: T is representable when it factors through a network of cost kernels whose boundary reduction is T , with the chord bound applying kernel by kernel; path independence and the chord condition would then be the image of the general condition in the cost-based case.

Accessible coherence, characterized. Compute $K_{\mathcal{A}}$ for concrete classes: polynomial-time arbitrageurs against cones whose separation is hard, budgeted arbitrageurs against fee-bearing venues, latency-constrained arbitrageurs against cascading settlements. And characterize the mechanisms in which the stream of certificate payments funds the ongoing computation.

The equilibrium fee. Section 7 argues discipline, not equilibrium. In a flow model where most coefficients are zero and informed trades carry signal plus noise, does the fee that survives undercutting reproduce the universal threshold $\sigma\sqrt{2\log p}$ of the shrinkage literature? If so, the regularization constant that statistics tunes by cross-validation equals the adverse-selection cost of the data source.

Loopy fixed points. Under a concrete flow model, does arbitrage-corrected quoting converge, and to what?

Arbitrage depth as a capacity measure. The arbitrage depth of a trained model's loss landscape is computable; whether it tracks quantities learning theory names (sharpness, mode connectivity) is open.

Effective friction. Proposition 15 eliminates frictionless networks. With per-maker fees, what are the effective fees and liquidities of the reduced boundary maker in terms of the network's, and which fee-bearing classes are closed under elimination?

Books around binary events. The multimodal-quote reading of book holes (§3) is testable

against limit-order data around court rulings and FDA decisions, where bimodal implied densities are documented (Clark and Amen 2017).

References

- Abernethy, Jacob, Yiling Chen, and Jennifer Wortman Vaughan. 2013. "Efficient Market Making via Convex Optimization, and a Connection to Online Learning." *ACM Transactions on Economics and Computation* 1 (2): 12:1–39. <https://doi.org/10.1145/2465769.2465777>.
- Abernethy, Jacob, Rafael Frongillo, Xiaolong Li, and Jennifer Wortman Vaughan. 2014. "A General Volume-Parameterized Market Making Framework." *Proceedings of the 15th ACM Conference on Economics and Computation (EC)*, 413–30. <https://doi.org/10.1145/2600057.2602900>.
- Aji, Srinivas M., and Robert J. McEliece. 2000. "The Generalized Distributive Law." *IEEE Transactions on Information Theory* 46 (2): 325–43. <https://doi.org/10.1109/18.825794>.
- Angeris, Guillermo, Tarun Chitra, Theo Diamandis, Alex Evans, and Kshitij Kulkarni. 2024. "The Geometry of Constant Function Market Makers." *Proceedings of the 25th ACM Conference on Economics and Computation (EC)*, 732. <https://doi.org/10.1145/3670865.3673500>.
- Barbu, Adrian, and Nathan Lay. 2012. "An Introduction to Artificial Prediction Markets for Classification." *Journal of Machine Learning Research* 13: 2177–204. <https://arxiv.org/abs/1102.1465>.
- Barrieu, Pauline, and Nicole El Karoui. 2005. "Inf-Convolution of Risk Measures and Optimal Risk Transfer." *Finance and Stochastics* 9 (2): 269–98. <https://doi.org/10.1007/s00780-005-0152-0>.
- Bertsekas, Dimitri P. 1992. "Auction Algorithms for Network Flow Problems: A Tutorial Introduction." *Computational Optimization and Applications* 1 (1): 7–66. <https://doi.org/10.1007/BF00247653>.
- Beygelzimer, Alina, John Langford, and David M. Pennock. 2012. "Learning Performance of Prediction Markets with Kelly Bettors." *Proceedings of the 11th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. <https://arxiv.org/abs/1201.6655>.
- Bhaskara, Adithya, Rafael Frongillo, and Maneesha Papireddygar. 2023. *A General Theory of Liquidity Provisioning for Prediction Markets*. arXiv:2311.08725. <https://arxiv.org/abs/2311.08725>.
- Biais, Bruno, David Martimort, and Jean-Charles Rochet. 2000. "Competing Mechanisms in a Common Value Environment." *Econometrica* 68 (4): 799–837. <https://doi.org/10.1111/1468-0262.00138>.
- Brahma, Aseem, Mithun Chakraborty, Sanmay Das, Allen Lavoie, and Malik Magdon-Ismael.

2012. "A Bayesian Market Maker." *Proceedings of the 13th ACM Conference on Electronic Commerce (EC)*, 215–32. <https://doi.org/10.1145/2229012.2229031>.
- Chen, Yiling, Lance Fortnow, Nicolas Lambert, David M. Pennock, and Jennifer Wortman. 2008. "Complexity of Combinatorial Market Makers." *Proceedings of the 9th ACM Conference on Electronic Commerce (EC)*, 190–99. <https://doi.org/10.1145/1386790.1386822>.
- Chen, Yiling, Sharad Goel, and David M. Pennock. 2008. "Pricing Combinatorial Markets for Tournaments." *Proceedings of the 40th ACM Symposium on Theory of Computing (STOC)*, 305–14. <https://doi.org/10.1145/1374376.1374421>.
- Chen, Yiling, and Jennifer Wortman Vaughan. 2010. "A New Understanding of Prediction Markets via No-Regret Learning." *Proceedings of the 11th ACM Conference on Electronic Commerce (EC)*, 189–98. <https://doi.org/10.1145/1807342.1807372>.
- Clark, Iain J., and Saeed Amen. 2017. "Implied Distributions from GBPUSD Risk-Reversals and Implication for Brexit Scenarios." *Risks* 5 (3): 35. <https://doi.org/10.3390/risks5030035>.
- Cummings, Rachel, David M. Pennock, and Jennifer Wortman Vaughan. 2016. "The Possibilities and Limitations of Private Prediction Markets." *Proceedings of the 17th ACM Conference on Economics and Computation (EC)*. <https://arxiv.org/abs/1602.07362>.
- d'Aspremont, Alexandre. 2005. *A Market Test for the Positivity of Arrow-Debreu Prices*. arXiv:cs/0510027. <https://arxiv.org/abs/cs/0510027>.
- Deza, Michel M., and Monique Laurent. 1997. *Geometry of Cuts and Metrics*. Vol. 15. Algorithms and Combinatorics. Springer. <https://doi.org/10.1007/978-3-642-04295-9>.
- Dudík, Miroslav, Xintong Wang, David M. Pennock, and David M. Rothschild. 2021. "Log-Time Prediction Markets for Interval Securities." *Proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 465–73. <https://doi.org/10.65109/bbzi6501>.
- Dudík, Miroslav, Sébastien Lahaie, and David M. Pennock. 2012. "A Tractable Combinatorial Market Maker Using Constraint Generation." *Proceedings of the 13th ACM Conference on Electronic Commerce (EC)*, 459–76. <https://doi.org/10.1145/2229012.2229047>.
- El Ghaoui, Laurent, and Hervé Lebrete. 1997. "Robust Solutions to Least-Squares Problems with Uncertain Data." *SIAM Journal on Matrix Analysis and Applications* 18 (4): 1035–64. <https://doi.org/10.1137/S0895479896298130>.
- Feigenbaum, Joan, and Scott Shenker. 2002. "Distributed Algorithmic Mechanism Design: Recent Results and Future Directions." *Proceedings of the 6th International Workshop on Discrete Algorithms and Methods for Mobile Computing and Communications (DIALM)*, 1–13. <https://www.cs.yale.edu/homes/jf/FS.pdf>.

- Frongillo, Rafael M., Nicolás Della Penna, and Mark D. Reid. 2012. "Interpreting Prediction Markets: A Stochastic Approach." *Advances in Neural Information Processing Systems 25 (NeurIPS)*. <https://papers.nips.cc/paper/4529-interpreting-prediction-markets-a-stochastic-approach>.
- Frongillo, Rafael, Maneesha Papireddygar, and Bo Waggoner. 2024. "An Axiomatic Characterization of CFMMs and Equivalence to Prediction Markets." *15th Innovations in Theoretical Computer Science Conference (ITCS)*, LIPIcs, vol. 287: 51:1–21. <https://doi.org/10.4230/LIPIcs.ITCS.2024.51>.
- Frongillo, Rafael, and Bo Waggoner. 2018. "Bounded-Loss Private Prediction Markets." *Advances in Neural Information Processing Systems (NeurIPS)*. <https://arxiv.org/abs/1703.00899>.
- Glosten, Lawrence R. 1994. "Is the Electronic Open Limit Order Book Inevitable?" *The Journal of Finance* 49 (4): 1127–61. <https://doi.org/10.1111/j.1540-6261.1994.tb02450.x>.
- Glosten, Lawrence R., and Paul R. Milgrom. 1985. "Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders." *Journal of Financial Economics* 14 (1): 71–100.
- Grone, Robert, Charles R. Johnson, Eduardo M. Sá, and Henry Wolkowicz. 1984. "Positive Definite Completions of Partial Hermitian Matrices." *Linear Algebra and Its Applications* 58: 109–24. [https://doi.org/10.1016/0024-3795\(84\)90207-6](https://doi.org/10.1016/0024-3795(84)90207-6).
- Guasoni, Paolo. 2006. "No Arbitrage Under Transaction Costs, with Fractional Brownian Motion and Beyond." *Mathematical Finance* 16 (3): 569–82. <https://doi.org/10.1111/j.1467-9965.2006.00283.x>.
- Guasoni, Paolo, Miklós Rásonyi, and Walter Schachermayer. 2010. "The Fundamental Theorem of Asset Pricing for Continuous Processes Under Small Transaction Costs." *Annals of Finance* 6 (2): 157–91. <https://doi.org/10.1007/s10436-008-0110-x>.
- Hanson, Robin. 2003. "Combinatorial Information Market Design." *Information Systems Frontiers* 5 (1): 107–19. <https://doi.org/10.1023/A:1022058209073>.
- Hanson, Robin. 2007. "Logarithmic Market Scoring Rules for Modular Combinatorial Information Aggregation." *The Journal of Prediction Markets* 1 (1): 3–15. <https://doi.org/10.5750/jpjm.v1i1.417>.
- Jouini, Elyès, and Hédi Kallal. 1995. "Martingales and Arbitrage in Securities Markets with Transaction Costs." *Journal of Economic Theory* 66 (1): 178–97. <https://doi.org/10.1006/jeth.1995.1037>.
- Kalman, Rudolf E. 1960. "A New Approach to Linear Filtering and Prediction Problems." *Journal of Basic Engineering* 82 (1): 35–45. <https://doi.org/10.1115/1.3662552>.

- Kalodner, Harry, Steven Goldfeder, Xiaoqi Chen, S. Matthew Weinberg, and Edward W. Felten. 2018. "Arbitrum: Scalable, Private Smart Contracts." *Proceedings of the 27th USENIX Security Symposium*, 1353–70. <https://www.usenix.org/conference/usenixsecurity18/presentation/kalodner>.
- Kroer, Christian, Miroslav Dudík, Sébastien Lahaie, and Sivaraman Balakrishnan. 2016. "Arbitrage-Free Combinatorial Market Making via Integer Programming." *Proceedings of the 17th ACM Conference on Economics and Computation (EC)*, 161–78. <https://arxiv.org/abs/1606.02825>.
- Lépinette, Emmanuel, and Tuan Quoc Tran. 2017. "Arbitrage Theory for Non Convex Financial Market Models." *Stochastic Processes and Their Applications* 127 (10): 3331–53. <https://www.sciencedirect.com/science/article/pii/S0304414915301216>.
- Li, Xiaolong, and Jennifer Wortman Vaughan. 2013. "An Axiomatic Characterization of Adaptive-Liquidity Market Makers." *Proceedings of the 14th ACM Conference on Electronic Commerce (EC)*, 657–74. <https://doi.org/10.1145/2482540.2482575>.
- Litvinov, Grigory L. 2005. *The Maslov Dequantization, Idempotent and Tropical Mathematics: A Brief Introduction*. arXiv:math/0507014. <https://arxiv.org/abs/math/0507014>.
- Marshall, Alfred. 1890. *Principles of Economics*. Macmillan.
- Mas-Colell, Andreu, Michael D. Whinston, and Jerry R. Green. 1995. *Microeconomic Theory*. Oxford University Press.
- Melick, William R., and Charles P. Thomas. 1997. "Recovering an Asset's Implied PDF from Option Prices: An Application to Crude Oil During the Gulf Crisis." *Journal of Financial and Quantitative Analysis* 32 (1): 91–115. <https://doi.org/10.2307/2331318>.
- Nau, Robert F., and Kevin F. McCardle. 1991. "Arbitrage, Rationality, and Equilibrium." *Theory and Decision* 31 (2–3): 199–240.
- Nisan, Noam, and Amir Ronen. 2001. "Algorithmic Mechanism Design." *Games and Economic Behavior* 35 (1–2): 166–96. <https://doi.org/10.1006/game.1999.0790>.
- Nisan, Noam, and Ilya Segal. 2006. "The Communication Requirements of Efficient Allocations and Supporting Prices." *Journal of Economic Theory* 129 (1): 192–224. <https://doi.org/10.1016/j.jet.2004.10.007>.
- Nueve, Enrique, Mary Nguyen, Rafael Frongillo, and Bo Waggoner. 2026. *Adaptive Liquidity in Prediction Markets via Online Learning*. arXiv:2605.09599. <https://arxiv.org/abs/2605.09599>.
- Nueve, Enrique, and Bo Waggoner. 2025. *Smooth Quadratic Prediction Markets*. arXiv:2505.02959. <https://arxiv.org/abs/2505.02959>.

- Olivares-Nadal, Alba V., and Victor DeMiguel. 2018. "A Robust Perspective on Transaction Costs in Portfolio Optimization." *Operations Research* 66 (3). <https://doi.org/10.1287/opre.2017.1699>.
- Othman, Abraham, David M. Pennock, Daniel M. Reeves, and Tuomas Sandholm. 2013. "A Practical Liquidity-Sensitive Automated Market Maker." *ACM Transactions on Economics and Computation* 1 (3). <https://doi.org/10.1145/2509413.2509414>.
- Othman, Abraham, and Tuomas Sandholm. 2011. "Liquidity-Sensitive Automated Market Makers via Homogeneous Risk Measures." *Internet and Network Economics (WINE 2011)*, Lecture notes in computer science, vol. 7090. https://doi.org/10.1007/978-3-642-25510-6_27.
- Othman, Abraham, and Tuomas Sandholm. 2012. "Profit-Charging Market Makers with Bounded Loss, Vanishing Bid/Ask Spreads, and Unlimited Market Depth." *Proceedings of the 13th ACM Conference on Electronic Commerce (EC)*, 790–807. <https://doi.org/10.1145/2229012.2229074>.
- Pearl, Judea. 1988. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann.
- Pennock, David M., and Michael P. Wellman. 1996. "Toward a Market Model for Bayesian Inference." *Proceedings of the 12th Conference on Uncertainty in Artificial Intelligence (UAI)*, 405–13. <https://arxiv.org/abs/1302.3593>.
- Pennock, David M., and Michael P. Wellman. 1997. "Representing Aggregate Belief Through the Competitive Equilibrium of a Securities Market." *Proceedings of the 13th Conference on Uncertainty in Artificial Intelligence (UAI)*, 392–400. <https://arxiv.org/abs/1302.1564>.
- Pennock, David M., and Michael P. Wellman. 2000. "Compact Securities Markets for Pareto Optimal Reallocation of Risk." *Proceedings of the 16th Conference on Uncertainty in Artificial Intelligence (UAI)*, 481–88. <https://arxiv.org/abs/1301.3886>.
- Rassenti, Stephen J., Vernon L. Smith, and Robert L. Bulfin. 1982. "A Combinatorial Auction Mechanism for Airport Time Slot Allocation." *The Bell Journal of Economics* 13 (2): 402–17. <https://doi.org/10.2307/3003463>.
- Rockafellar, R. Tyrrell. 1970. *Convex Analysis*. Princeton Mathematical Series 28. Princeton University Press. <https://doi.org/10.1515/9781400873173>.
- Saguillo, Oriol, Vahid Ghafouri, Lucianna Kiffer, and Guillermo Suarez-Tangil. 2025. "Unraveling the Probabilistic Forest: Arbitrage in Prediction Markets." *Advances in Financial Technologies (AFT)*. <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.AFT.2025.27>.
- Schweppe, Fred C., Michael C. Caramanis, Richard D. Tabors, and Roger E. Bohn. 1988. *Spot Pricing of Electricity*. Kluwer Academic Publishers.

- Storkey, Amos J. 2011. "Machine Learning Markets." *Proceedings of the 14th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, vol. 15: 716–24. <https://arxiv.org/abs/1106.4509>.
- Storkey, Amos J., Jono Millin, and Krzysztof Geras. 2012. "Isoelastic Agents and Wealth Updates in Machine Learning Markets." *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/1206.6443>.
- Sun, Wei, Robin Hanson, Kathryn B. Laskey, and Charles Twardy. 2012. "Probability and Asset Updating Using Bayesian Networks for Combinatorial Prediction Markets." *Proceedings of the 28th Conference on Uncertainty in Artificial Intelligence (UAI)*, 815–24. <https://arxiv.org/abs/1210.4900>.
- Teutsch, Jason, and Christian Reitwießner. 2019. *A Scalable Verification Solution for Blockchains*. arXiv:1908.04756. <https://arxiv.org/abs/1908.04756>.
- Wellman, Michael P. 1993. "A Market-Oriented Programming Environment and Its Application to Distributed Multicommodity Flow Problems." *Journal of Artificial Intelligence Research* 1: 1–23. <https://doi.org/10.1613/jair.2>.
- Xia, Lirong, and David M. Pennock. 2011. "Price Updating in Combinatorial Prediction Markets with Bayesian Networks." *Proceedings of the 27th Conference on Uncertainty in Artificial Intelligence (UAI)*, 825–34. <https://arxiv.org/abs/1202.3756>.
- Xu, Huan, Constantine Caramanis, and Shie Mannor. 2009. "Robustness and Regularization of Support Vector Machines." *Journal of Machine Learning Research* 10: 1485–510. <https://jmlr.org/papers/v10/xu09b.html>.